

Graphs - Set 1

Generalities. Incidence and adjacency matrices

1. Let $G = (V, E)$ be a simple graph with n vertices, with $\deg(v) \geq \frac{n-1}{2}$ for all $v \in V$. Prove that G is connected.
2. For all the digraphs in Figures 1-4, write the incidence and adjacency matrices.
3. Which of the digraphs in Figure 1 are strongly connected?

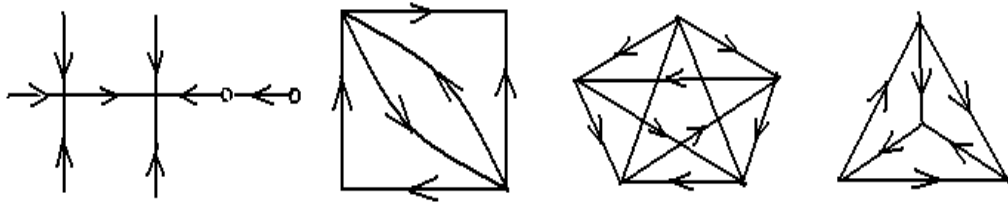


Figure 1: Problem 3.

4. In the digraph in Figure 2, find
 - a. all the cycles of length 3, 4 and 5,
 - b. an Eulerian trail, and
 - c. a Hamiltonian cycle.

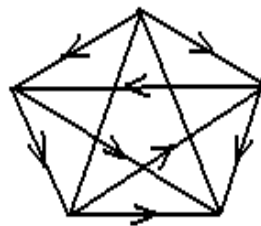


Figure 2: Problem 4.

5. In the digraph in Figure 3 give (if possible)
 - a. a walk of length 7 from u to w
 - b. cycles of length 1, 2, 3 and 4 and
 - c. a path of maximum length.
6. Design an algorithm analogous to "Basic tree-growing algorithm with vertex labeling" that grows a tree in a digraph. What should be the definition of a *frontier arc*? Apply this algorithm for the graph in Figure 4, starting from the following vertices:
 - a. vertex u

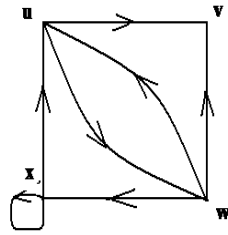


Figure 3: Problem 5.

- b. vertex x
- c. vertex w
- d. vertex y .

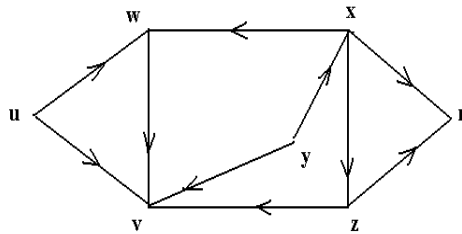


Figure 4: Problem 6.

7. Prove that two graphs G_1, G_2 , having the adjacency matrices A_1, A_2 , are isomorphic if and only if there exists a permutation matrix P such that $A_2 = P^{-1}A_1P$.